

Terraform EC2 Module

File Structure

```
modules/
  ■■■ ec2/
    ■■■ main.tf
    ■■■ variables.tf
    ■■■ outputs.tf
    ■■■ user_data/
      ■■■ user_data.sh
      ■■■ app_user_data.sh
```

main.tf

```
resource "aws_instance" "web" {
  count = length(var.public_subnet_ids)

  ami           = var.ami_id
  instance_type = var.instance_type
  subnet_id     = var.public_subnet_ids[count.index]
  key_name      = var.key_name
  vpc_security_group_ids = [var.web_security_group_id]

  user_data = file("${path.module}/user_data/user_data.sh")

  tags = {
    Name = "web-server-${count.index + 1}"
    Tier = "Web"
  }
}

resource "aws_instance" "app" {
  count = length(var.private_app_subnet_ids)

  ami           = var.ami_id
  instance_type = var.instance_type
  subnet_id     = var.private_app_subnet_ids[count.index]
  key_name      = var.key_name
  vpc_security_group_ids = [var.app_security_group_id]

  user_data = file("${path.module}/user_data/app_user_data.sh")

  tags = {
    Name = "app-server-${count.index + 1}"
    Tier = "Application"
  }
}
```

variables.tf

```
variable "vpc_id" {
  type = string
}

variable "public_subnet_ids" {
  type = list(string)
}

variable "private_app_subnet_ids" {
  type = list(string)
}

variable "instance_type" {
```

```

    type = string
}

variable "ami_id" {
    type = string
}

variable "key_name" {
    type = string
}

variable "web_security_group_id" {
    type = string
}

variable "app_security_group_id" {
    type = string
}

```

outputs.tf

```

output "web_instance_ids" {
    value = aws_instance.web[*].id
}

output "app_instance_ids" {
    value = aws_instance.app[*].id
}

output "web_public_ips" {
    value = aws_instance.web[*].public_ip
}

output "app_private_ips" {
    value = aws_instance.app[*].private_ip
}

```

user_data/user_data.sh

```

#!/bin/bash

apt-get update -y
apt-get install -y nginx

systemctl enable nginx
systemctl start nginx

cat > /var/www/html/index.html <<'EOF'
<!DOCTYPE html>
<html>
<head>
    <title>3-Tier AWS Application</title>
</head>
<body>
    <h1>3-Tier AWS Infrastructure</h1>
    <p>Nginx Web Server is running successfully.</p>
</body>
</html>
EOF

```

user_data/app_user_data.sh

```

#!/bin/bash

```

```
apt-get update -y
apt-get install -y apache2 php php-mysql

systemctl enable apache2
systemctl start apache2

mkdir -p /var/www/html

cat > /var/www/html/index.php <<'EOF'
<?php
echo "<h1>Application Tier</h1>";
echo "<p>Private Application Server is running.</p>";
?>
EOF
```